

UNIT TEST (GRADE XII)

F.M. : 25

P.M. : 10

SUB: CHEMISTRY

TIME: 1 HOUR

Attempt all Questions

[5 × 5 = 25]

1. An aliphatic haloalkane (A) is heated with aq. KOH to give (B). The compound (B) reacts with $SOCl_2$ to produce (C). The compound (C) further reacts with $AgCN$ to yield (D). On reduction of compound (D) with $LiAlH_4$ produces (E). The compound (A) undergoes Wurtz reaction to give an alkane of molecular formula C_4H_{10} . Identify (A), (B), (C), (D) and (E) with reactions involved. [5]
2. Write the various reactions inside the blast furnace during extraction of iron. What happens when iron is treated with *conc.* HNO_3 ? [4 + 1]
3. (a) How $CHCl_3$ can be prepared from (i) Ethanol (ii) Acetone?
(b) What happens when chloroform is (i) treated with *conc.* HNO_3 (ii) reduced in acidic medium (iii) heated with aqueous alkali? [2 + 3]
4. (a) Derive expression of Ostwald's dilution law for weak electrolyte.
(b) Write limitation of Ostwald's dilution law.
(c) Define the terms (i) Weak electrolyte (ii) Degree of dissociation. [2 + 1 + 2]
5. (a) Define the terms (i) Decinormal solution (ii) Primary standard substance.
(b) It is asked to prepare a decinormal solution of Na_2CO_3 in 100 mL volumetric flask and the solution is prepared by dissolving 0.636 g Na_2CO_3 . Find the normality factor of the prepared decinormal solution. Also calculate the volume of water required to make the prepared solution exactly decinormal. [2 + 3]

Best of Luck

UNIT TEST (GRADE XII)

F.M. : 25

P.M. : 10

SUB: PHYSICS

TIME: 1 HOUR

1. a) State and prove Ampere's circuital law? [3]
b) A toroid has a core of inner radius of 20 cm and outer radius of 25 cm. The core has a windings of 1500 turns of the wire and current of 2A was passed through the wire. Find the magnetic flux density inside and outside toroid. [2]
2. a) Define cross field? Mention its use in JJ Thompson experiment. [2]
b) Two plane metal plates 4.0 cm long are held horizontally 3.0 cm apart in a vacuum, one being vertically above the other. The upper plate is at a potential of 300 V and the lower is earthed. Electrons having a velocity of $1.0 \times 10^7 \text{ ms}^{-1}$ are injected horizontally midway between the plates and in a direction parallel to the 4.0 cm edge. Calculate the vertical deflection of the electron beam as it emerges from the plates. [e/m for electron = $1.8 \times 10^{11} \text{ Ckg}^{-1}$] [3]
3. a) Describe Millikan's experiment to calculate the value of Planck's constant. [3]
b) The photoelectric work function of potassium is $3.8 \times 10^{-19} \text{ J}$. If light of wavelength $5000 \text{ \AA}$ is incident on it, find (i) the stopping potential (ii) the kinetic energy of the most energetic electrons and (iii) the velocity of these electrons. [2]
4. A tuning fork vibrates under an external periodic force.
a) What is resonance? Under what condition does it occur?
b) What is the phase relation between displacement and driving force at resonance?
c) A forced oscillator has natural frequency $\omega_0 = 5 \text{ rad/s}$. When driven by an external force of frequency $\omega = 5 \text{ rad/s}$, amplitude is maximum. Explain this behavior using a graph and calculate the resonance condition.
5. a) Describe Laplace's correction to Newton's formula for the velocity of sound in gases. How does considering adiabatic change instead of isothermal ones correct the result? [3]
b) A gas has adiabatic index (γ) = 1.4 and molar mass 28.97 g/mol . Calculate the speed of sound in this gas at 300 K. (Use $R = 8.314 \text{ J/mol. K}$) [2]

BEST OF LUCK

UNIT TEST (CLASS XII)

SUBJECT: MATHEMATICS
TIME: 1 HOUR

FULL MARKS: 25

PASS MARKS: 10

GROUP A

1. a) What are elementary row operations? [1]
b) What are the two major two steps of Gauss elimination method? [1]
c) State addition principle of counting. [1]
d) What is the value of integral $\int \frac{1}{\sqrt{a^2-t^2}} dt$? [1]
e) Define order and degree of differential equations. [1]

GROUP B

2. a) Test the consistency of the following system by using Gauss elimination method.
$$2x - y = 5$$
$$x - \frac{1}{2}y = \frac{5}{2}$$
 [2]
b) In how many ways can odd numbers of three different digits be formed by using 0, 5, 6, 7, 8, 9? [3]
3. a) Evaluate: $\int \frac{x^4-1}{x^2+4} dx$ [2]
b) For what values of x , the rate of increase of $y = x^3 - 5x^2 + 5x + 8$ is twice the rate of increase of x ? [3]
4. a) Define proper rational fraction with an example. Under what condition it becomes proper? [2]
b) Under what condition, the system of linear equations of two variables becomes consistent? Justify your answer with an example. [3]
5. a) Evaluate: $\int \frac{1}{x(x^n+1)} dx$ [3]
b) By using determinant method, solve $\frac{2}{x} - \frac{3}{y} = 5$ and $\frac{1}{x} + \frac{4}{y} = 7$. [2]

Best of Luck